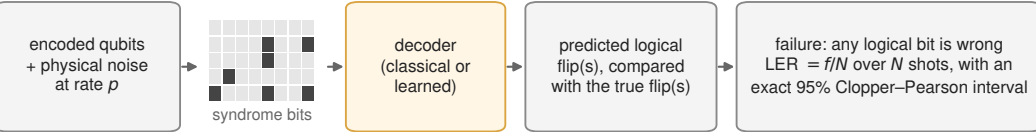


The decoding task



Three evaluation settings, each with its own reference decoder (never compared across settings)

<i>code</i>	<i>noise model</i>	<i>decoder input</i>	<i>reference decoder</i>
A rotated surface code $d = 5, 7, 9, 11$	circuit level: gates, measurements and resets can fail	25-round detector history	MWPM (PyMatching)
B bivariate-bicycle (BB) codes [[72, 12, 6]], [[144, 12, 12]]	code capacity: data flips only, perfect checks, one round	one syndrome	BP-OSD (ldpc)
C BB codes [[72, 12, 6]] ($T = 6$) and [[144, 12, 12]] ($T = 12$)	phenomenological: data and measurement flips over T rounds	T -round detector history	BP-OSD on the space-time checks

Protocol: failure count and exact interval for every LER · paired McNemar test on the same shots · single-frame call latency with intervals